

Chess Games Analysis · 66,879 Games

SQL Data Cleaning, Parsing & Analysis Case Study — SQLite + Power BI

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1. PROJECT OVERVIEW

This project analyzes 66,879 Chess.com club games entirely in SQL to answer four questions: does White win more often, which openings perform best, does the higher-rated player win, and do results change by time control. The emphasis is on doing the real work — cleaning, text parsing, feature engineering, and aggregation — in SQL, with Power BI reserved as the visualization layer. The goal was a portfolio piece that demonstrates practical, interview-relevant SQL rather than surface-level querying.

2. DATASET

Attribute	Detail
Source	Kaggle — Chess.com club games (club_games_data.csv)
Rows (games)	66,879 (65,778 standard chess after scope filter)
Columns	14
Key raw fields	white_rating, black_rating, white_result, time_class, rules, rated, pgn
Buried in PGN	opening name, ECO code, termination — extracted via string parsing

3. APPROACH & TOOLS

All data preparation and analysis were performed in SQLite using DB Browser. The raw CSV was imported as a table (games), all cleaning and feature engineering were consolidated into a single reusable view (games_clean), and each analytical question was answered with a short query against that view. Power BI connects to the cleaned data as the visualization layer — the analytical logic lives in SQL, not in the BI tool.

4. DATA ASSESSMENT

The dataset had **no missing values**, but the most valuable information was locked inside the free-text pgn column — the opening name, ECO code, and termination were embedded in a block of tags rather than sitting in their own fields. Extracting them was the core cleaning challenge. Standard chess accounts for 65,778 of 66,879 games (98.4%); the remainder are variants (crazyhouse, king-of-the-hill, three-check, chess960) that follow different rules.

5. DATA CLEANING & FEATURE ENGINEERING (SQL)

All logic was consolidated into a single view. Key decisions:

- **Scope — standard chess only:** filtered rules = 'chess', excluding ~1.6% of non-comparable variants.
- **Derived outcome:** a CASE expression converts White's result code into White win / Black win / Draw.
- **Parsed opening & ECO:** extracted from the PGN using INSTR + SUBSTR string functions.
- **Rating gap:** rating_diff = white – black, wrapped in CAST because SQLite imported all columns as text.

The clean view

```
CREATE VIEW games_clean AS
SELECT white_username, black_username, white_rating, black_rating,
       time_class, rated,
       CASE
         WHEN white_result = 'win' THEN 'White win'
         WHEN white_result IN ('resigned','checkmated','timeout','abandoned') THEN 'Black win'
         ELSE 'Draw'
       END AS outcome,
       REPLACE(SUBSTR(SUBSTR(pgn, INSTR(pgn,'openings/')+9), 1,
         INSTR(SUBSTR(pgn, INSTR(pgn,'openings/')+9), '"') - 1), '-', ' ') AS opening,
       SUBSTR(pgn, INSTR(pgn,'[ECO ')+6, 3) AS eco,
       CAST(white_rating AS INTEGER) - CAST(black_rating AS INTEGER) AS rating_diff
```

```
FROM games
WHERE rules = 'chess';
```

6. SQL TECHNIQUES DEMONSTRATED

- **Text parsing** — INSTR + SUBSTR to extract opening and ECO from raw PGN, including nested SUBSTR to locate the correct closing delimiter.
- **Feature engineering** — CASE expressions to derive outcome and rating-gap buckets.
- **Subquery** — dividing each result count by a total computed in a nested SELECT.
- **Aggregation & filtering** — GROUP BY with HAVING to keep only openings with a reliable sample size.
- **Proportion via AVG(CASE)** — turning wins into 1/0 and averaging to get win rate.
- **Views** — consolidating all derived logic into a reusable staging layer.

7. ANALYSIS & FINDINGS

Q1 — Win rate by color

Outcome	Games	%
White win	32,783	49.84%
Black win	30,697	46.67%
Draw	2,298	3.49%

Finding: White wins measurably more than Black — a clear first-move advantage of roughly 3 percentage points. Draws are rare in club-level play (3.5%).

Q2 — Openings and win rate

Finding (highest White win rate, min. 50 games): the top openings by White win rate include King's Pawn Opening (88.0% over 249 games), Petrov's Defense Classical Damiano Variation (69.6%), Nimzowitsch-Larsen Attack (67.4%), Sicilian Defense McDonnell Attack (65.2%), and Queen's Pawn Opening (65.1%).

Analyst note: King's Pawn Opening's 88% win rate is an outlier well above the ~50% baseline and should be read with caution — it most likely reflects very short, low-level games rather than a genuinely dominant line. Flagging this rather than reporting it at face value is part of honest analysis.

Most-played openings (by count): Bishop's Opening (1,191), Scandinavian Defense (993), Queen's Pawn Opening Accelerated London System (959), Sicilian Defense Bowdler Attack (848), and Van't Kruijs Opening (815) lead the list — a mix of solid mainstream and offbeat club-level choices.

Q3 — Does the higher-rated player win?

Rating gap bucket	Games	White win %	Black win %
Black +100 or more	6,545	17.2%	80.2%
Black slightly higher	25,873	33.3%	62.8%
White slightly higher / even	26,808	66.1%	30.4%
White +100 or more	6,552	81.3%	16.0%

Finding: rating is a strong, clean predictor of the result. White's win rate climbs monotonically from 17% (when 100+ points lower) to 81% (when 100+ points higher), and Black's win rate mirrors it almost symmetrically. Most games are between closely-matched players (the two middle buckets hold ~79% of games).

Q4 — Results by time control

Time control	Games	White win %	Black win %	White edge
Blitz	28,805	49.9%	46.1%	+3.8
Bullet	22,222	50.1%	47.6%	+2.5
Rapid	12,872	49.3%	45.9%	+3.4
Daily	1,879	49.1%	48.4%	+0.7

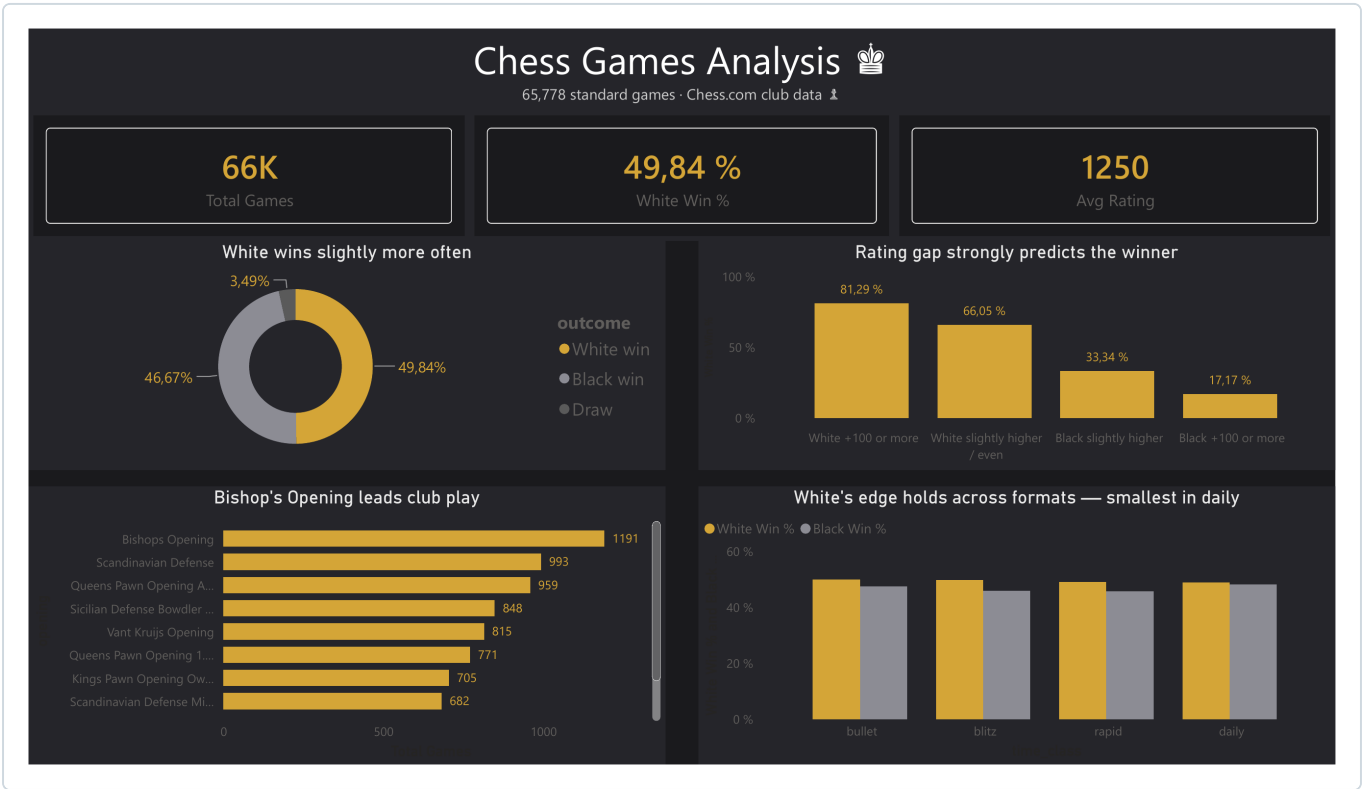
Finding: White's win rate is remarkably stable (~49-50%) across all formats, so the first-move advantage holds regardless of pace. The interesting nuance is the *size* of the edge: it is largest in blitz (+3.8 pts) and nearly disappears in daily / correspondence chess (+0.7 pts) — with days to think per move, Black neutralizes the opening advantage far more effectively.

8. KEY INSIGHTS

- White holds a real first-move advantage (49.8% vs 46.7% among decisive games).
- Rating gap is the dominant driver of results — win rate ranges from ~17% to ~81% across the rating spectrum, near-symmetric between colors.
- Opening win rates vary, but extreme values (e.g., King's Pawn at 88%) likely reflect game length and player skill rather than opening strength — treated as a caution, not a headline.
- White's advantage is consistent across time controls but shrinks with thinking time — strong in blitz (+3.8 pts) and almost gone in daily play (+0.7 pts).

9. VISUALIZATION LAYER (POWER BI)

The cleaned view `games_clean` feeds a Power BI dashboard with a dark, professional theme (custom JSON theme, amber accent). It connects to the SQLite database via ODBC — the analytical logic stays in SQL, while Power BI serves as the presentation layer. KPI cards summarize scale, and each of the four analyses is presented with an intentional chart type: a donut for the part-to-whole color split, columns for the rating-gap progression, horizontal bars for openings, and grouped columns for time control.



Chess Games Analysis — Power BI dashboard (SQLite via ODBC)

10. CHALLENGES & LESSONS LEARNED

- **Parsing free text in SQL.** The opening name lived inside the PGN. A first attempt matched an earlier "] from another tag; the fix was a nested SUBSTR so the search for the closing delimiter began only after openings/.
- **SQLite types.** DB Browser imports every column as text, so numeric operations (`rating_diff`) required CAST to INTEGER.
- **GROUP BY discipline.** Every column in a grouped SELECT must be either grouped or aggregated — raw columns like `rating_diff` cannot sit loose in the SELECT.
- **Reproducibility.** Consolidating all derived logic into a view kept the analysis queries short and made the whole pipeline easy to re-run and hand off.

11. LIMITATIONS

Analysis is limited to standard chess (variants excluded). Win rates for very rare openings are unreliable, which is why a minimum sample threshold was applied; even so, some outliers likely reflect game length or skill rather than the opening itself. The dataset captures club games and is not necessarily representative of all rating ranges or of professional play.

Repository contents: chess_analysis.sql (commented queries + view), this documentation, and the Power BI dashboard. **SQL used:** CASE, INSTR, SUBSTR, REPLACE, CAST, COUNT, AVG, GROUP BY, HAVING, subquery, CREATE VIEW.